

"Dylan Soth Resume"

Dylan Soth

Mechanical Engineer with 2+ years of professional experience developing healthcare technology. Demonstrates a passion for finding solutions that improve patient care and quality of life. Seeking opportunities to leverage can-do attitude in developing cutting-edge medical technologies that address critical healthcare challenges.

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Education

B.S. in Mechanical Engineering, Northeastern University (2019-2023)

- University Honors Program Distinction, 3.81/4.0 GPA
- Labwork in Measurements, Mechanics, and Anatomy and Physiology.
- Activities with NU ASME, NU Robotics Club, and NU Service-Learning.
- Elective coursework in Flight, Film, and Global Health.

Experience

Certified TMS Technician, River City TMS, PLLC (2023-present, Spokane, WA)

- **Patient Care:** Administered hundreds of transcranial magnetic stimulation (TMS) treatments to dozens of patients who have suffered MDD.
- **Quality Improvement:** Designed a training program; led 4 hands-on demonstration hours, each with a prepared protocol study/SOP.
- **Informatics:** Wrote a Python script to grab hundreds of patients' data and calculate new variables for each treatment.
- **Research:** Discovered and implemented novel methods for targeting treatment sites with direction from medical staff.

R&D Electromechanical Co-op, Siemens Healthineers Endovascular Robotics (July 2022 - December 2022, Newton, MA)

- **Test and Integration:** Assembled eight Alpha prototype systems. Used test methods to verify system performance at key breakpoints in assembly.
- **Accelerated Life Testing:** Scripted CMM programs to quantify gear profiles. Designed fixture for thermal chamber testing.

- **Procurement:** Managed quote generation, fulfillment, and tracking for dozens of unique components. .
- **PDM:** Used SolidWorks PDM to grab dozens of drawings and models for procurement and rapid prototyping.

Assistant Research Engineer Co-op, Liberating Technologies, Inc (July 2021 - December 2021, Holliston, MA)

- **Verification Testing:** Led test stand operation and plotted exponential decay in grasp force for a gear-driven prosthesis.
- **Work Instructions:** Developed IMU-based system calibration protocol. Iterated embedded hardware validation using oscilloscope and software tests.
- **Embedded Systems:** Implemented in-house API and SDK libraries in C to control PWM motors. Completed a feature branch pull request and merge while practicing Git commit best practices

Skills

Lab Tools: Strain Gauges, Encoders, IMUs; Mitutoyo CMM, SLA/FDM 3D Printers, Wet Lab PPE

Software: SolidWorks PDM (CSWP), MATLAB, Arduino, LabVIEW, Unity, git

Projects

Assistive Mini-Golf System, Senior Capstone (2022-2023)

- **Design:** Designed and developed an assistive device enabling a quadriplegic client to play mini-golf. Implemented precise aiming and power control using stepper motors, timing belts, and pulleys.
- **Testing:** Successfully tested the system with the client, achieving target hitting and bank shots within five minutes of play.

Notes

Programming: Python, MATLAB, Arduino, SEGGER Embedded Studio, git, AutoCAD, ANSYS Workbench,